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Space-based Proton Electron Energy Detector (SPEED)—Simulation, Data Acquisition and Readout Electronics

A Thesis

Submitted by

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Of

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THESIS CERTIFICATE

This is to undertake that the Thesis titled **SPACE-BASED PROTON ELECTRON ENERGY DETECTOR (SPEED)SIMULATION, DATA ACQUISITION AND READOUT ELECTRONICS**, submitted by me to the Indian Institute of Technology Madras, for the award of **Bachelor of Technology**, is a bona fide record of the research work done by me under the supervision of **Dr. David Koilpillai**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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This work was highly interdisciplinary in nature and required learning across several technical domains that were entirely new to me at the start. I am grateful to everyone involved in IITMSAT for giving me the opportunity to learn and grow with such remarkable freedom.

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ABSTRACT

KEYWORDS IITMSAT; SPEED; Muons; Geant4; CAEN; Scintillator; Low Earth Orbit; charged particle flux; microcontroller; TIVA; photomultiplier tubes; ADC; DMA; timers; energy binning; space radiation

The Space-based Proton Electron Energy Detector (SPEED) is the primary scientific payload of IITMSAT, a student-built satellite at IIT Madras, designed to measure the flux and energy spectrum of charged particles precipitating from the Van Allen radiation belts in Low Earth Orbit. This thesis presents contributions across three areas of payload development. In simulation, the Geant4 detector framework was extended and used to characterise energy deposition from cosmic muons and radioactive sources, with additional noise modelling studies quantifying the effect of ambient gamma radiation and WLS fibre reflectivity on detector response. In data acquisition, cosmic muon calibration data was collected using the CAEN MC² Analyser and analysed using Gaussian and Landau fitting in Python. In readout electronics, critical firmware bugs were identified and resolved alongside a full firmware audit for port assignments and signal gain. This work collectively brings the SPEED payload significantly closer to flight readiness.

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GLOSSARY

Analog-to-Digital Converter	A peripheral that samples an analogue voltage and converts it to a discrete digital value for processing by the microcontroller.
Direct Memory Access	A hardware feature that transfers data between peripherals and memory without CPU involvement, used to transmit energy histogram data every three seconds.
Interrupt Service Routine	A firmware function executed automatically when a hardware interrupt fires, handling time-sensitive events such as timer expiry.
Landau Distribution	An asymmetric distribution describing energy loss fluctuations of a fast charged particle in a thin absorber, with a sharp peak and long high-energy tail.
Monte Carlo Simulation	A statistical technique that simulates large numbers of individual particle histories to model detector response.
Particle Burst	A sudden sharp increase in the count rate of high-energy protons or electrons in the upper ionosphere.
Peak-Hold Detector	An analogue circuit that captures and holds the maximum voltage of a pulse for sampling by the ADC.
Photomultiplier Tube	A vacuum tube that converts incident photons to an electrical signal via the photoelectric effect and dynode multiplication.
Physics List	A Geant4 configuration that specifies

which physical processes are active in a simulation.

Scintillator

A material that emits light proportional to the energy deposited by an incident charged particle.

Stopping Power

The rate of energy loss per unit length of a charged particle in matter, $\frac{dE}{dx}$, used for particle identification via the Bethe-Bloch formula.

Synchronous Serial Interface

A clocked serial communication protocol used to transmit science packets from the payload microcontroller to the onboard computer.

Universal Asynchronous Receiver-Transmitter An asynchronous serial protocol used for ground testing and debug verification of science packet contents.

Van Allen Radiation Belts

Two toroidal regions of energetic charged particles trapped by Earth's magnetic field in the inner magnetosphere.

Wavelength Shifting Fibre

An optical fibre that absorbs UV scintillation photons and re-emits them at a longer visible wavelength, guiding them to the PMT.

ABBREVIATIONS

ADC	Analog-to-Digital Converter.
DMA	Direct Memory Access.
FEE	Front-End Electronics.
GPIO	General Purpose Input Output.
HV	High Voltage.
IITMSAT	IIT Madras Student Satellite.
ISR	Interrupt Service Routine.
LEO	Low Earth Orbit.
PMT	Photomultiplier Tube.
SPEED	Space-based Proton Electron Energy Detector.
SSI	Synchronous Serial Interface.
UART	Universal Asynchronous Receiver-Transmitter.
WLS	Wavelength Shifting.

CHAPTER 1

INTRODUCTION

The IITMSAT is a student-built satellite that has been designed around a central scientific instrument SPEED (Space-based Proton Electron Energy Detector). SPEED is a scintillator based charged particle detector designed to measure the energy and flux of protons and electrons in the upper ionosphere.

1.1 SPEED PAYLOAD

The Space-based Proton Electron Energy Detector is a scintillator detector system for measuring the energy spectrum, background flux, and fluctuations in the flux of protons and electrons in the upper ionosphere. The planned study made using this detector will help us validate and refine existing trapped particle radiation models and analyze radiation belt dynamics.

Space radiation environments are highly dynamic due to solar flares and geomagnetic fluctuations. Precise data on these phenomenon helps modeling radiation belt behaviour, which directly impacts the safety and efficiency for Low Earth Orbits (LEO) missions and systems. Research fields like solar physics, ionospheric studies, and electromagnetism rely on this data for high-fidelity modeling.

1.2 OBJECTIVES

The primary objective of the study is to analyse the nature of fluctuations of charged particle fluxes in Low Earth Orbit due to their precipitation from the Van-Allen radiation belts by measuring the position and time dependent behaviours. We are interested in three specific measurements. They are as follows:

- **Particle Energy Spectrum:** Map how particle energies are distributed across the orbit and characterise the proton and electron energy ranges encountered at altitude.
- **Background Flux:** Measure the steady-state count rate of high-energy particles in the upper ionosphere as a function of orbital position.
- **Particle Bursts:** Identify and characterise sudden, sharp increases in proton/electron flux as well as the nature, duration, and recurrence of such events.

The secondary objective is to experimentally characterize the relationship between particle bursts (sharp increase in count rates of protons and electrons) in the upper ionosphere

with space and earth weather phenomena such as lightning storms, geomagnetic storms, solar flares and potentially even seismic activity. This is an active area of research and the data from SPEED could make a genuine scientific contribution to it.

1.3 THREE AREAS OF CONTRIBUTIONS

My contributions to this project fall into three distinct technical areas:

1. Geant4 Simulations
2. Data Acquisition and Analysis
3. Readout Electronics

All three areas are deeply related to each other and are collectively necessary to thoroughly understand the working of the payload, SPEED. The simulation work informed the experimental strategy, the experimental data validated the simulation models, and the firmware is what will ultimately bring all of this together to collect science data. Chapters 3 to 5 cover these contributions in detail.

Chapter 3 goes into the updates made to the Geant4 simulations of the detector for multiple sources. Geant4 is a Monte Carlo particle transport toolkit developed at CERN, and it serves as the primary tool for understanding how different particles interact with the SPEED detector geometry before any physical measurements are made. Simulations were performed for cosmic muons and radioactive sources, with additional studies conducted to model realistic noise conditions expected in orbit.

Chapter 4 discusses the data acquisition methods and tools used, as well as the data analysis in Python that followed. Calibration measurements were performed using the laboratory replica of the SPEED payload, and the collected data was processed and analysed to extract meaningful detector characterisation parameters.

Chapter 5 details the back-end digital electronics that enable science signal readout and energy binning. The microcontroller firmware is responsible for reading, sorting, and transmitting science data to the onboard computer, and many contributions related to maintaining the correctness and functionality of this pipeline.

CHAPTER 2

SPEED

2.1 ABOUT SPEED

The Space-based Proton Electron Energy Detector (SPEED) is a scintillator-based charged particle detector, composed of multiple scintillator layers for energy measurement and particle identification. Specifically, SPEED is designed to measure the energy spectrum, background flux, and fluctuations in proton and electron flux in upper ionosphere.

IITMSAT is particularly designed to operate in Low Earth Orbits (LEO) which range from 200 to 2000 km above sea levels. Satellites in LEO experience significant radiation from the Van Allen belts, particularly when passing through the South Atlantic Anomaly (a region where the inner belt dips unusually close to Earth's surface).

The Van Allen Radiation belts refer to two donut-shaped regions of energetic charged particles (protons and electrons) trapped by Earth's magnetic field. The inner belt is dominated by high-energy protons, the outer by electrons. SPEED is designed to measure these particles as IITMSAT passes through them in Low Earth Orbit.

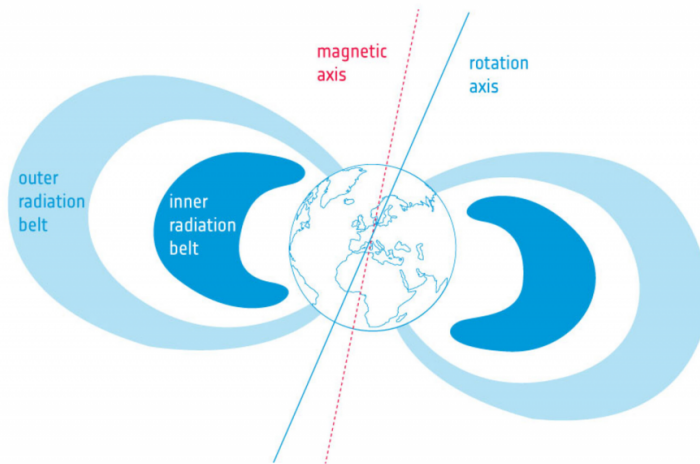


Figure 2.1: Illustration showing Van Allen belts. Credits: ESA

2.2 STRUCTURE OF SPEED

SPEED is designed with the satellite constraints in mind, which dictate the size and placement of its various components. The detector consists of three distinct scintillator layers

- the dEdx layer
- the Bulk layer
- the Veto layer

each serving a specific and complementary role in measuring and identifying incident charged particles.

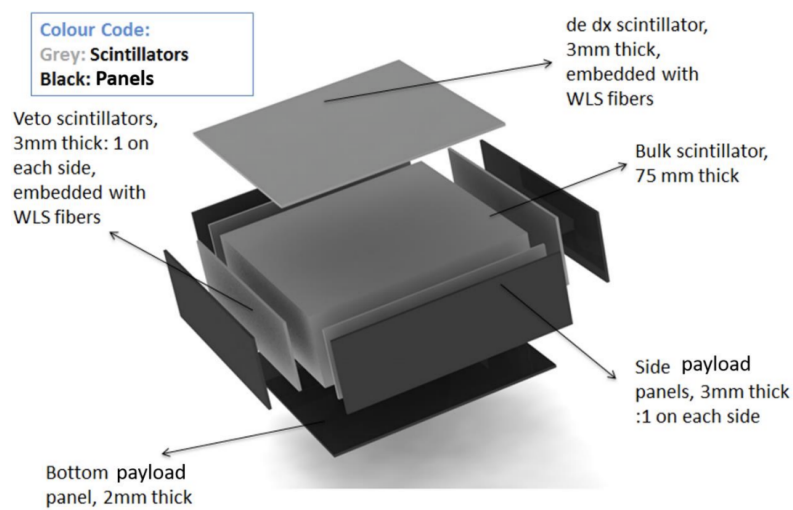


Figure 2.2: Scintillator configuration of SPEED. Credits: IITMSAT

The dEdx scintillator, 3 mm thick, forms the top layer of the detector. It is intentionally kept thin so that an incident particle passes through it while depositing only a fraction of its total energy, rather than stopping inside it. This partial energy deposit is used to measure the rate of energy loss per unit length, $\frac{dE}{dx}$, which is a characteristic property of the particle described by the Bethe-Bloch formula. Since $\frac{dE}{dx}$ depends on the charge and velocity of the particle, comparing the measured energy loss against the particle's total energy allows us to distinguish between particle species — in particular, between protons and electrons at the same kinetic energy, which have very different stopping power profiles.

Directly beneath the dEdx layer sits the bulk scintillator, which at 75 mm thick forms the core of the detector. Its large volume is designed to completely stop and absorb particles that pass through the dEdx layer, capturing their remaining kinetic energy. The sum of the energy deposited in the dEdx layer and the energy deposited in the bulk together gives the total initial kinetic energy of the incident particle. It is this two-layer coincidence measurement that makes SPEED capable of both energy spectroscopy and

particle identification simultaneously.

The veto scintillators line all four sides of the detector and serve a data quality function rather than a measurement function. Any particle that triggers a veto scintillator is flagged and discarded from the science data. This covers two scenarios — a particle entering the detector from the side rather than through the dEdx layer, or a particle that passes straight through the bulk without stopping and therefore does not deposit its full energy. In either case the two-layer energy sum would be incomplete and the measurement unreliable. The veto system ensures that only well-contained, fully characterised events contribute to the final science data, which is essential for the integrity of the energy spectrum measurements.

Wavelength Shifting (WLS) optical fibres are placed in grooves made in the dEdx and Bulk layers. The scintillators are then wrapped using Mylar and two layers of black Kapton using 3M Sheets. This ensures the maximum photon exit from the fibres to the photomultiplier tubes (PMTs).



Figure 2.3: WLS fibre arrangement for dEdx. Credits: IITMSAT

2.3 WORKING OVERVIEW

When a charged particle enters the SPEED detector, it first goes through the dEdx scintillator, depositing a fraction of its energy and causing the material to emit scintillation light. The particle then continues into the bulk scintillator, where it is fully stopped and deposits its remaining kinetic energy. If the particle instead passes through the bulk or enters from the sides, the veto scintillators surrounding the detector register a signal, and the event is flagged and discarded.

The scintillation light produced in each layer is collected by wavelength shifting fibres embedded in the grooves within the scintillator material. These fibres absorb the scintillation photons and re-emit them at a longer visible wavelength, guiding them to the photomultiplier tubes. There are three PMTs in total with one dedicated to the dEdx layer, one to the bulk layer, and one shared across all veto panels. Each PMT converts the incoming photons into an electrical current pulse through the photoelectric effect and subsequent dynode multiplication, producing a pulse whose amplitude is proportional to the energy deposited in the corresponding scintillator layer.

This pulse is then processed by the front-end electronics, which amplify and shape the signal before a peak-hold detector captures its maximum amplitude. The held voltage which is proportional to the deposited energy is sampled by the ADC on the microcontroller, which sorts the reading into the appropriate energy bin of an accumulated histogram. Every three seconds, this histogram is packaged into a science packet and transmitted to the satellite's onboard computer for downlink to the ground station, where it is decoded to reconstruct the energy spectrum and flux of incident particles.

CHAPTER 3

GEANT4 SIMULATION

3.1 ABOUT GEANT4

Geant4 [Agostinelli *et al.* (2003); Geant4 Collaboration (2023)] is a Monte Carlo simulation toolkit developed and maintained by CERN, designed to simulate the passage of particles through matter. It is written in C++ using an object-oriented framework, which allows users to define custom detector geometries, materials, particle sources, and physics processes in a modular and flexible manner. Geant4 is the industry standard tool for detector simulation in high energy physics, nuclear physics, and space science, and is used extensively by experiments at CERN as well as by space agencies including NASA and ESA for radiation environment studies.



Figure 3.1: Geant4 Logo. Credits: CERN

Monte Carlo refers to the statistical approach that forms the basis of the toolkit. Instead of solving a single deterministic equation for how a particle traverses matter, Geant4 simulates a large number of individual particle histories. At each step along a particle's trajectory, the toolkit samples from detailed physics cross-section tables to probabilistically determine what interaction occurs like whether the particle scatters, ionises the material, decays, or produces secondary particles. By simulating thousands of such histories, the aggregate behaviour converges to physically accurate distributions of energy deposition, particle flux, and detector response.

Geant4 organises its physics modelling into physics lists which are configurable sets of processes that are active during a simulation. These include electromagnetic processes such as ionisation, Bremsstrahlung, and Compton scattering; hadronic processes; radioactive decay chains; and optical photon transport. Choosing the appropriate physics list for a given application is an important step, as it determines both the physical accuracy and the computational cost of the simulation. For the SPEED detector, the relevant processes are primarily electromagnetic interactions and radioactive decay, given the energy ranges and particle types of interest.

3.2 SPEED SIMULATIONS

The Geant4 simulation framework for SPEED models the full detector geometry, material composition, and optical properties of the scintillator system. The simulation is used

to study the response of the detector to different particle sources, characterise energy deposition patterns across the dEdx and bulk layers, and model the effect of various noise sources on the recorded signal. The results of these simulations helped in identifying key patterns and placement strategies for laboratory data collection and calibration.

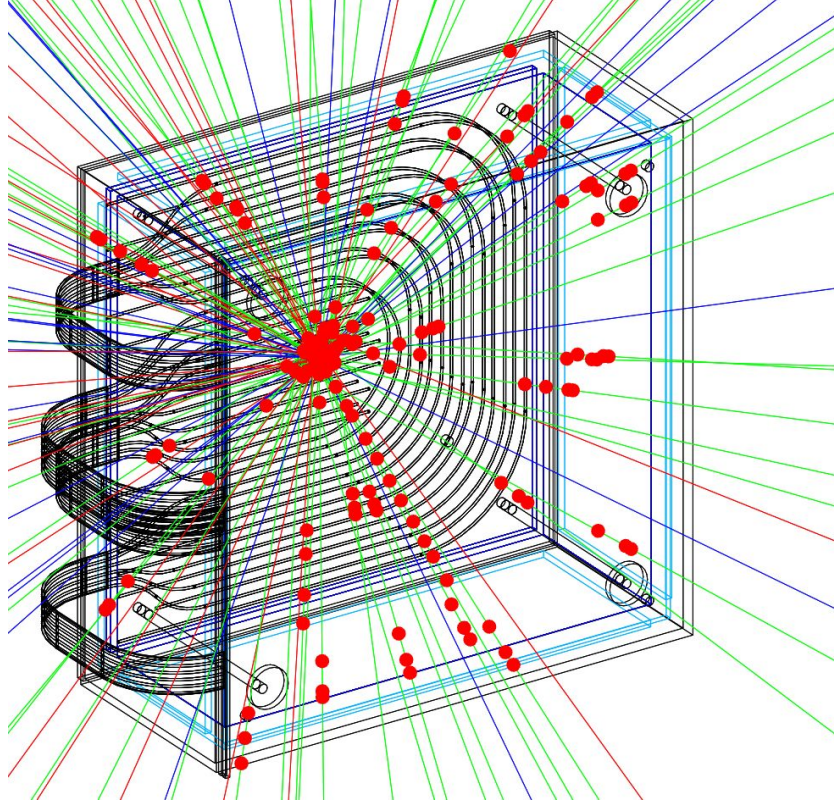


Figure 3.2: SPEED Simulation perpendicularly above the centre of dEdx

3.2.1 Detector Geometry Implementation

The SPEED detector geometry is constructed in Geant4 using the standard hierarchy of solid, logical, and physical volumes. The world volume is defined as a $250\text{ cm} \times 250\text{ cm} \times 250\text{ cm}$ air-filled box serving as the mother volume for all subsequent placements.

The scintillator material used for both the dEdx and bulk layers is BC444 which is a vinyltoluene-based plastic scintillator (C_9H_{10}) with a density of 1.032 g/cm^3 and implemented as a custom `G4Material` with Birks' constant set to account for ionisation quenching at high energy deposition densities. The veto scintillators use a separate instance of the same material, `BC444_2`, to allow independent sensitive detector assignment. Optical properties including refractive index, absorption length, and scintillation emission spectrum are assigned through `G4MaterialPropertiesTable` objects when optical physics is enabled.

The full detector assembly is enclosed within an aluminium casing (`AlCasing`), constructed as a `G4Box` solid and assigned as a sensitive detector volume to track any energy deposited in the casing material itself. Inside this casing is the scintillator

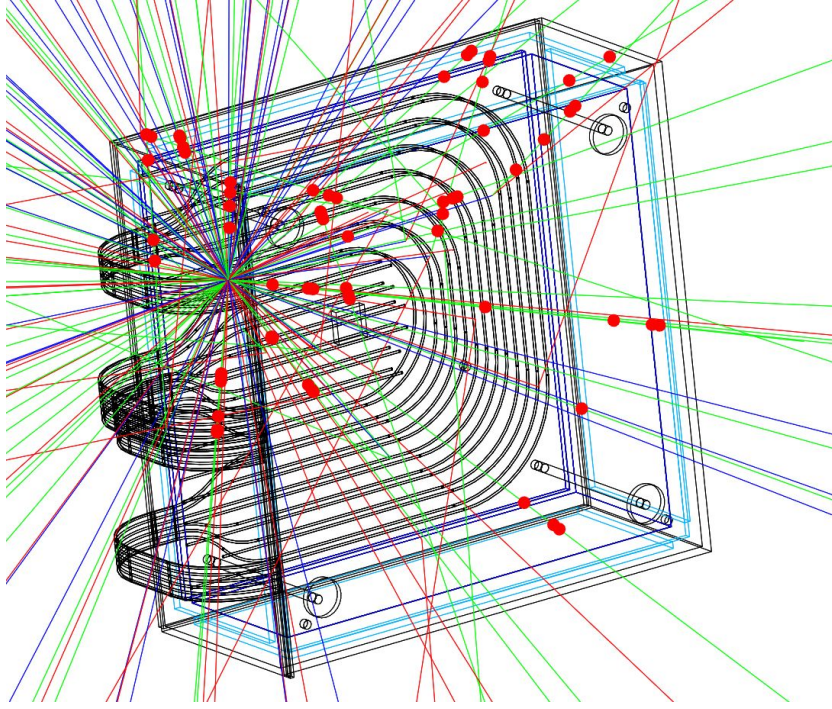


Figure 3.3: SPEED Simulation perpendicularly above the centre of dEdx with and offset

assembly volume (`SctAssembly`), which serves as the immediate mother volume for the individual scintillator layers. Each scintillator is surrounded by a layered wrapping stack consisting of black Kapton polyimide for light tightness, Mylar for reflective support, and a thin aluminised reflective layer. They are implemented as concentric `G4Box` volumes with aluminium surfaces assigned optical reflectivity of 0.85 via `G4OpticalSurface` and `G4LogicalSkinSurface`.

The wavelength shifting fibres are modelled as cylindrical `G4Tubs` volumes with a polystyrene core and PMMA cladding, with optical properties for the BCF-91A fibre assigned via material properties tables specifying the WLS absorption length spectrum, re-emission spectrum, attenuation length of 3.5 m, refractive index of 1.60 for the core and 1.49 for the cladding, and a time constant of 12 ns. The fibre routing geometry includes straight sections within the scintillator volume and curved sections modelled as `G4Torus` solids outside it, with the fibres bunched at the readout end to concentrate them towards the PMT window.

Each detector layer is registered as a separate sensitive detector using the `G4SDManager`, enabling independent hit collection and energy deposit recording for each layer during simulation runs.

3.2.2 Cosmic Muon Simulations

Cosmic muons are simulated using Geant4's General Particle Source (GPS) module. Both μ^+ and μ^- are included at a fixed kinetic energy of 100 MeV, which is representative of the lower end of the cosmic muon energy spectrum at sea level and ensures complete

stopping within the bulk scintillator for most trajectories.

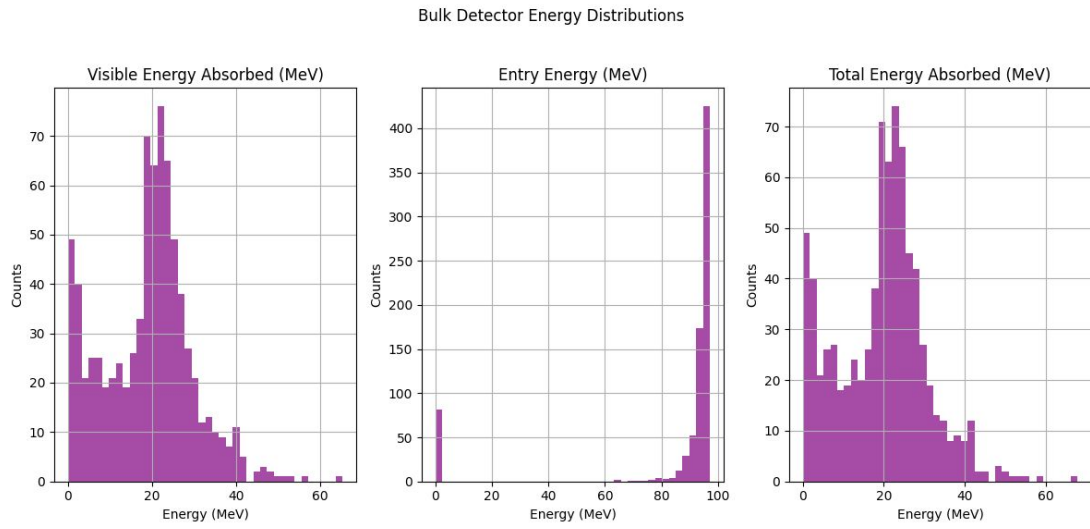


Figure 3.4: Bulk Detector Energy Distribution - Muons

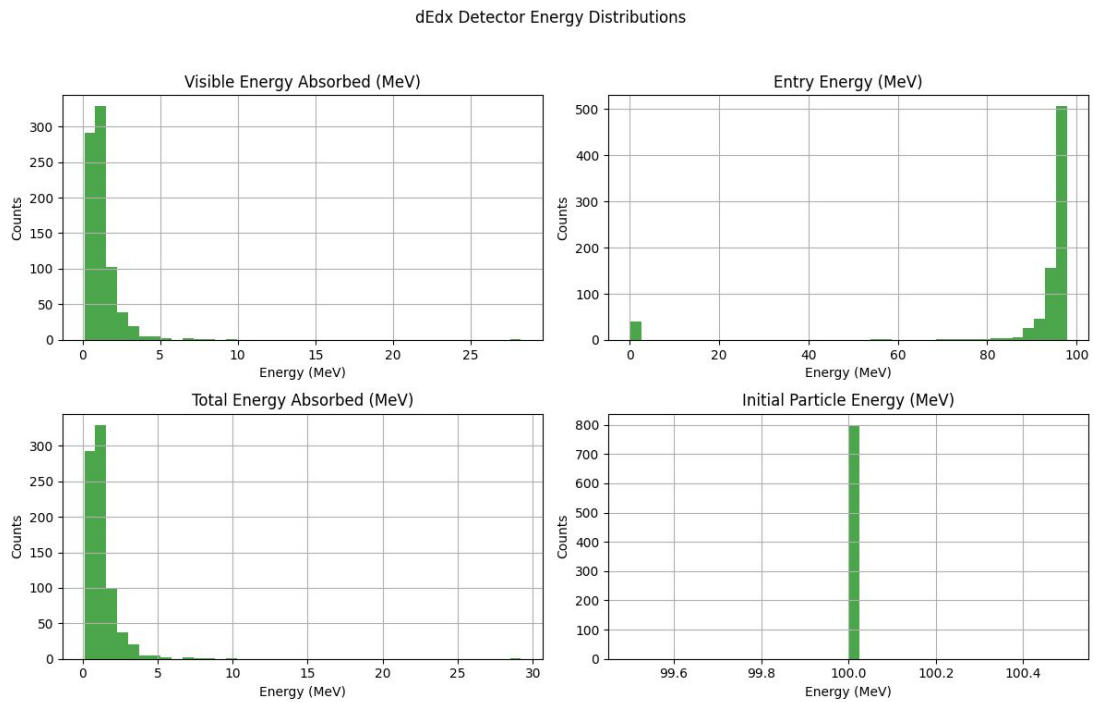


Figure 3.5: dEdx Detector Energy Distribution - Muons

The source geometry is defined as a planar square surface of half-width 100 mm in both transverse directions, positioned at $x = -2$ mm on top of the dEdx top surface, which is placed at the origin. The source plane is oriented such that its normal points along the positive x -axis, aligned with the detector's main particle acceptance direction, using $\text{rot1} = (0, 1, 0)$ and $\text{rot2} = (0, 0, 1)$ to define the plane.

The angular distribution is set to cosine-weighted ($\cos$) over a polar angle range of 0 to 90 from the beam normal, which replicates the $\cos^2 \theta$ angular distribution characteristic of cosmic muons at the Earth's surface. This ensures the simulated flux is forward-directed and physically realistic in its angular spread rather than being purely in one direction.

Trajectory visualisation is configured using the DAWNFILE driver with smooth trajectory rendering and hit accumulation across events, allowing the full muon track distribution through the detector geometry to be inspected post-run.

3.2.3 P-32 Radioactive Source Configuration

The P-32 calibration source is simulated using Geant4's GPS ion source interface, specified as `/gps/ion 15 32 0 0` which translates to atomic number 15 (phosphorus), mass number 32, with zero charge state and zero excitation energy. This instructs Geant4 to initialise the nucleus in its ground state and handle the subsequent radioactive decay chain internally through the RadioactiveDecay physics process, which samples the decay mode, decay products, and product energies from the evaluated nuclear data libraries included in the Geant4 data package.

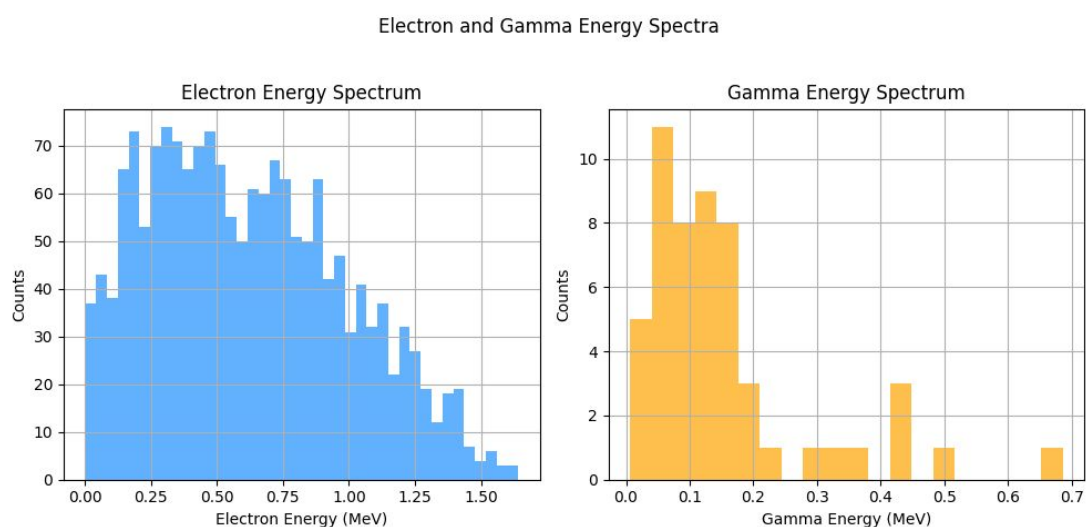


Figure 3.6: Electron and Gamma Energy Spectrum - P32

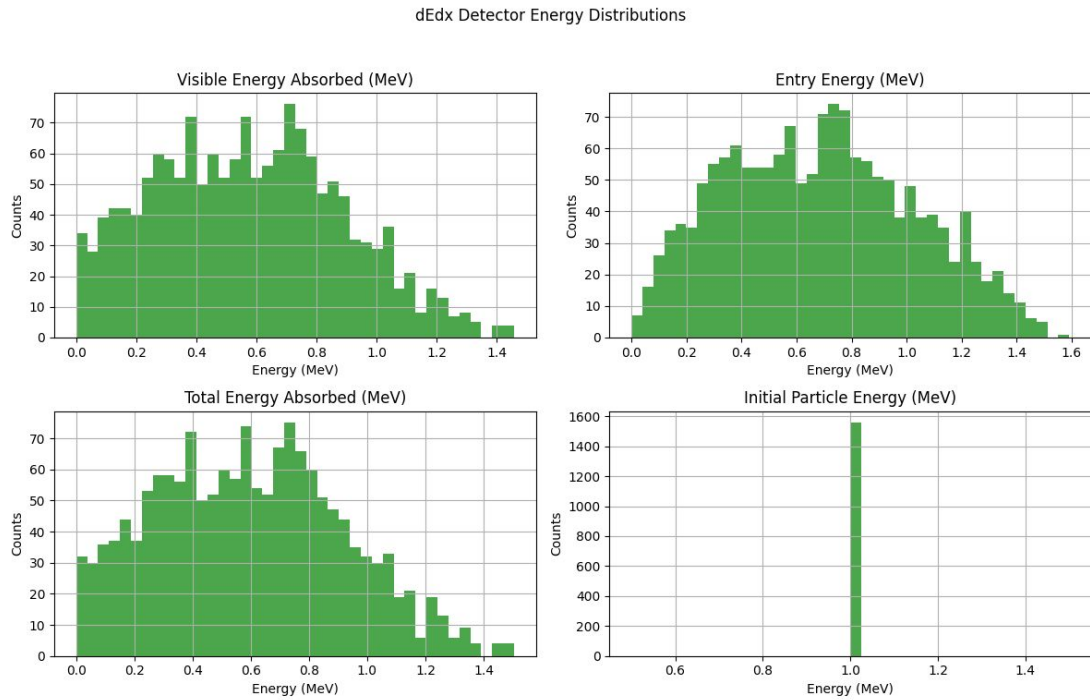


Figure 3.7: dEdx Detector Energy Distribution - P32

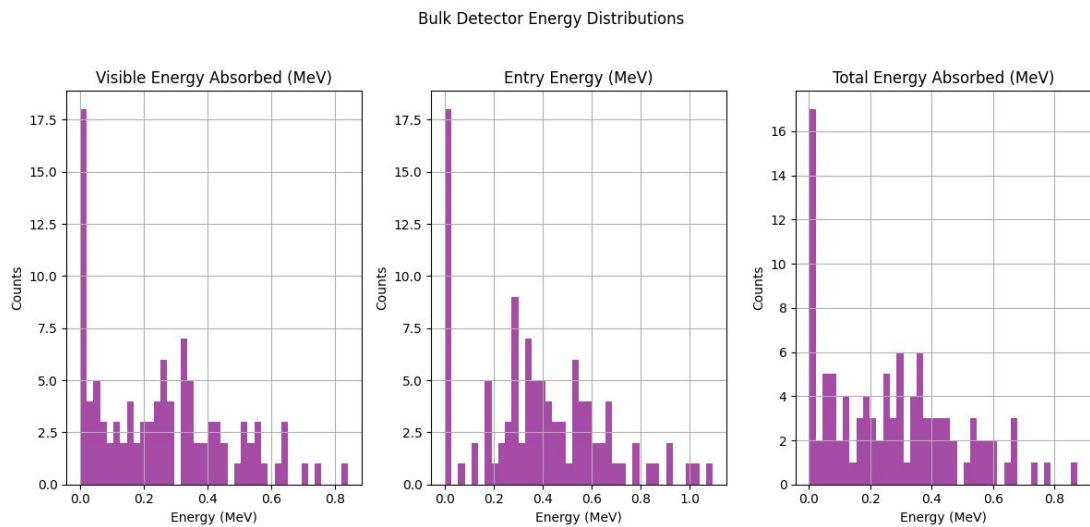


Figure 3.8: dEdx Detector Energy Distribution - P32

P-32 undergoes pure beta-minus decay with a half-life of 14.3 days, emitting an electron with a continuous energy spectrum up to a maximum endpoint energy of 1.71 MeV

and a mean energy of approximately 0.69 MeV, along with an antineutrino. No gamma emission occurs in the primary decay, making P-32 a clean beta source well suited for characterising the dEdx layer response to electrons in the sub-MeV range.

The source is placed as a point source at $x = -2$ mm which coincides with the muon source position, just on top of the dEdx surface with an isotropic angular distribution (iso). A total of 50000 decay events are simulated to accumulate sufficient statistics across the full beta energy spectrum, given that many low-energy electrons will stop in the source itself or the upstream air gap without reaching the detector.

3.2.4 Cs-137 Gamma Source Configuration

The Cs-137 configuration differs fundamentally from the P-32 run in both source type and physics. Rather than using the radioactive ion interface, it is configured as a direct monoenergetic gamma source at 662 keV which is the characteristic gamma energy from the de-excitation of the Ba-137m daughter nucleus following beta decay of Cs-137. This approach bypasses the full decay chain and isolates the gamma interaction physics, which is the primary quantity of interest for energy calibration using this source.

The 662 keV gamma is placed at the same point source location with an isotropic angular distribution and 5000 events are specified which is appropriate since gamma interactions are less frequent than direct beta ionisation and the photopeak statistics per event are higher. The Compton edge of this source appears at approximately 477 keV and the photopeak at 662 keV, providing two well-defined spectral features that can be used as absolute energy reference points for calibrating the bulk detector's ADC-to-energy relationship.

3.2.5 Noise Modelling

To study the effect of ambient gamma radiation on detector performance, the surrounding air volume of the world was populated with a gamma emitter at tunable emission rates. The emission rate was varied across four values: 10, 20, 50, and 100 gammas per second. This helped us to simulate a range of background radiation intensities representative of different orbital environments. At each rate, the resulting energy distributions in both the dEdx and bulk detectors were recorded and compared against the clean source signal.

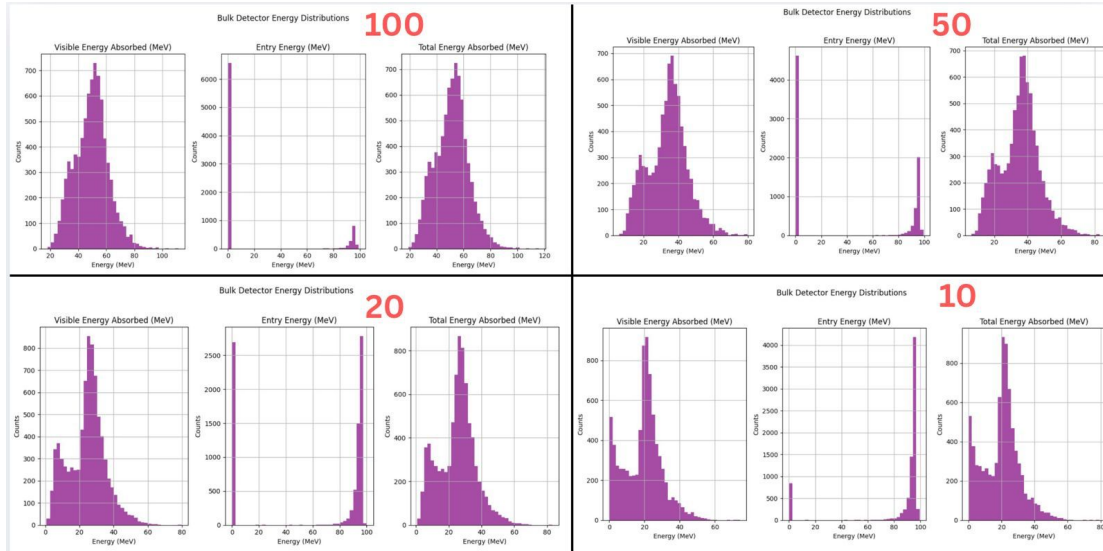


Figure 3.9: Bulk Energy Distribution with varying emissions rates (in red)

The second noise study investigated the sensitivity of light collection efficiency to the optical properties of the WLS fibres. The surface reflectivity of the fibre wrapping was measured at two values: 0.1 and 0.5. The resulting change in the visible energy absorbed at the PMT sensitive surface directly quantified the dependence of the detector's signal amplitude on fibre reflectivity, providing guidance on the acceptable tolerance in fibre surface quality during detector assembly.

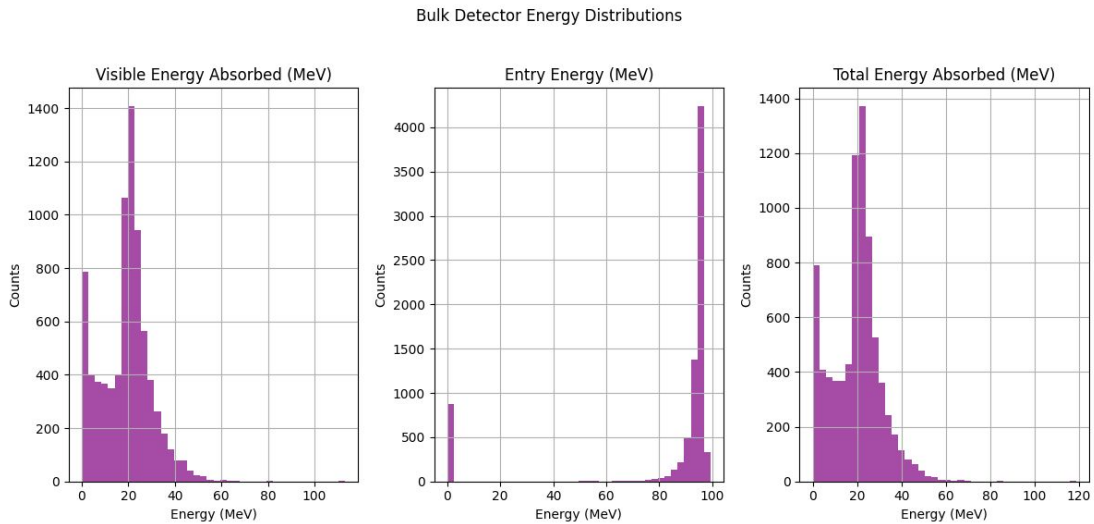


Figure 3.10: Bulk Energy Distribution with WLS Reflectivity = 0.1

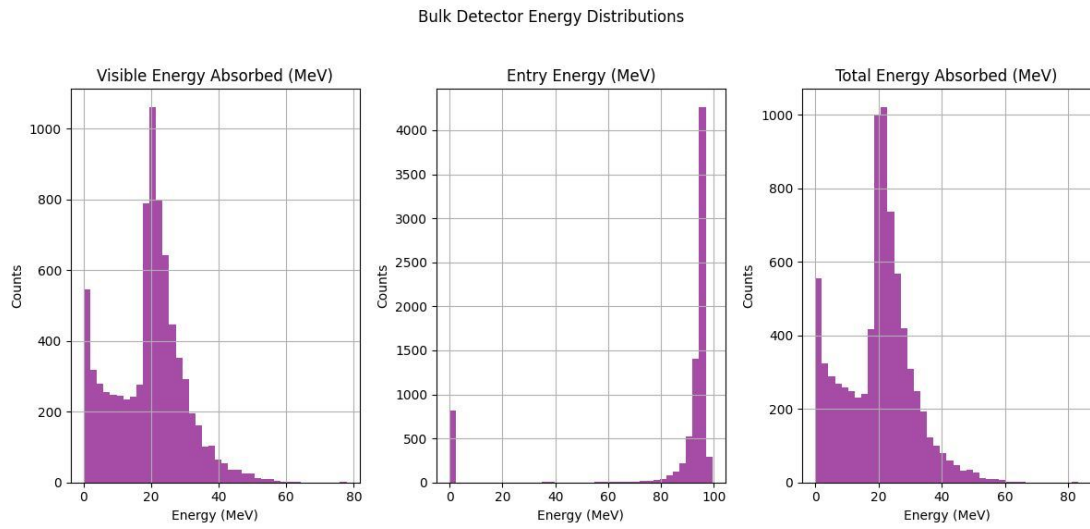


Figure 3.11: Bulk Energy Distribution with WLS Reflectivity = 0.5

CHAPTER 4

DATA ACQUISITION & ANALYSIS

4.1 OVERVIEW

Laboratory data acquisition was performed using the physical replica of the SPEED payload built with the same scintillator materials, wavelength shifting fibres, and photomultiplier tubes as the flight unit. This was done to collect calibration data that could be directly compared against the Geant4 simulation results, and to characterise the detector's response to known particle sources before the custom firmware readout chain was fully operational.

Rather than relying on the microcontroller-based science data pipeline, a commercially available CAEN MC² Analyser [CAEN S.p.A. (2020)] was used as an independent digitizer. This instrument connects directly to the analog output of the front-end electronics and provides a multi-channel amplitude histogram in real time, allowing rapid assessment of the detector's pulse amplitude distribution without requiring the full embedded firmware to be functional.

4.2 EXPERIMENTAL SETUP

The lab replica of the SPEED payload was configured with the front-end electronics connected to the PMT outputs of the dEdx and bulk scintillator channels. The CAEN MC² Analyser was connected to the peak-hold output of the front-end electronics and operated using the CAEN MC² Software, which provides a graphical interface for configuring acquisition parameters, setting discriminator thresholds, and recording amplitude histograms. Data was collected for two source configurations: cosmic muons using the detector in its natural radiation environment, and a P-32 beta source placed at the dEdx surface. Both runs accumulated over a sufficient time window to collect statistically meaningful histograms.

4.3 COSMIC MUON

Cosmic muons were used as the primary source due to their well-understood [Workman *et al.* (2022)] and highly penetrating nature. As minimum ionising particles at the energies encountered at sea level, cosmic muons deposit a characteristic and relatively uniform amount of energy as they traverse the full detector stack, producing a recognisable signal in the bulk channel. This property made them useful for verifying the detector geometry, confirming coincidence logic, and providing an absolute energy reference.

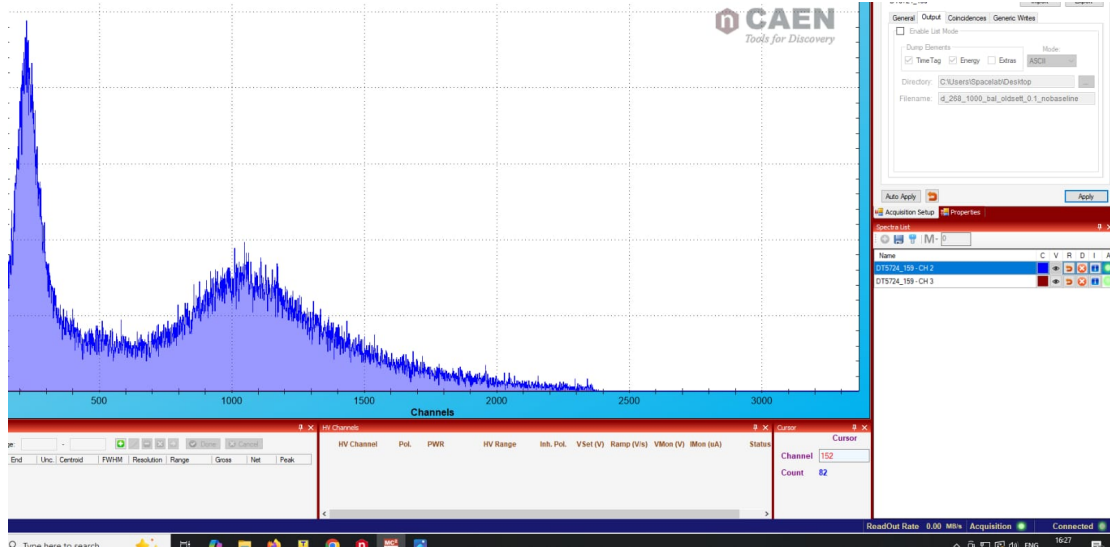


Figure 4.1: Data Collected in CAEN MC² Software on Lab Replica Model for Cosmic Muons

4.4 DATA ANALYSIS

The raw amplitude histogram collected by the CAEN system contains contributions from two components, firstly a low-amplitude noise peak arising from electronic baseline fluctuations and dark current pulses from the PMT, and secondly a broader higher-amplitude distribution corresponding to genuine muon energy deposits, which follows a Landau distribution characteristic of energy loss by minimum ionising particles in matter. The analysis was performed in Python using a three-step procedure.

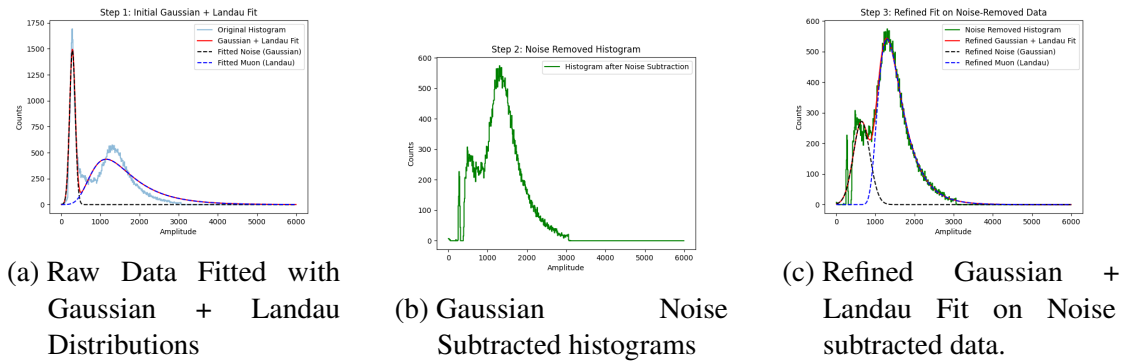


Figure 4.2: Data Collected in CAEN MC² Software on Lab Replica Model for Cosmic Muons

In the first step, the raw histogram was fitted with a combined Gaussian plus Landau distribution. The Gaussian component captures the noise peak at low amplitude, while

the Landau component which is characterised by its asymmetric shape with a sharp rising edge and a long high-energy tail was used for the muon signal. This simultaneous fit allows the two components to be cleanly separated even in the region where they overlap. In the second step, the fitted Gaussian noise component was subtracted from the raw histogram to isolate the muon signal. In the third step, the noise-subtracted data was fitted again with Gaussian plus Landau Distributions to obtain the final muon spectrum. The final selection was used to extract the necessary timing details.

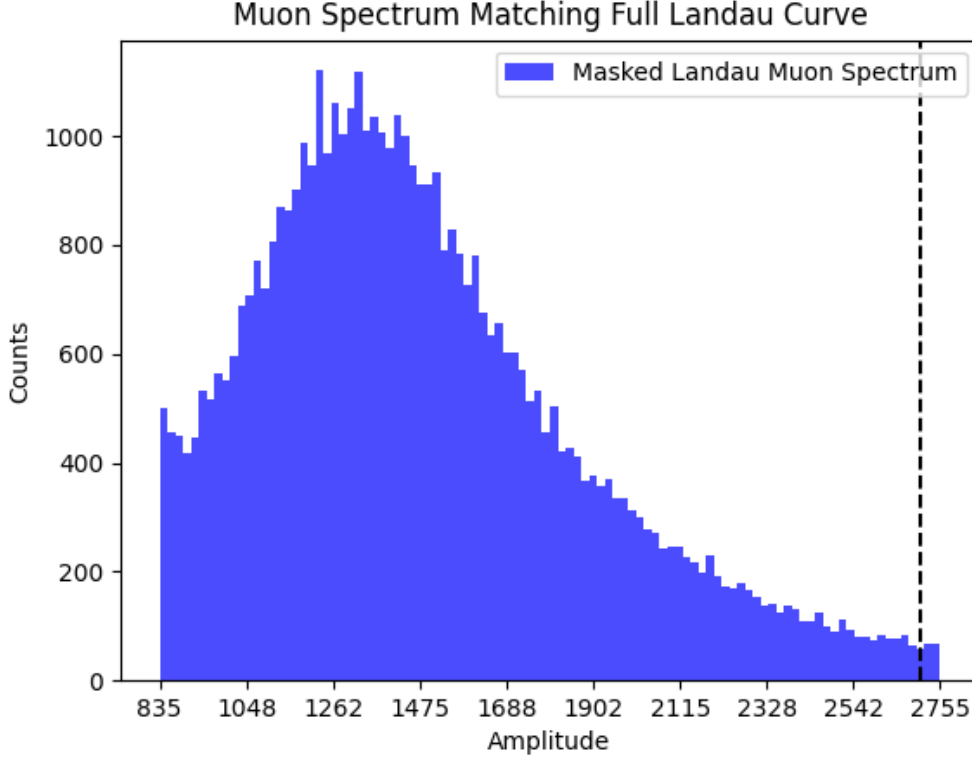


Figure 4.3: Final Muon Spectrum after three step selection and noise subtraction.

The inter-event time differences recorded in the acquisition output which were analysed to extract the detector's timing characteristics. Time differences greater than 600 time units (each unit corresponding to 10 ns) were excluded as unphysical outliers, and the minimum time offset was subtracted to reference the distribution to zero. The resulting time difference histogram was fitted with an exponential decay function

$$N(t) = N_0 \exp\left(-\frac{t}{\tau}\right) \quad (4.1)$$

using `scipy.optimize.curve_fit`, fitting only bins with non-zero counts to avoid bias from empty bins at large time separations. The fitted decay constant τ characterises

the mean inter-arrival time of muon events and is consistent with the expected cosmic muon flux at sea level given the detector's active area. This exponential distribution of inter-arrival times is the expected statistical behaviour for a Poisson process, confirming that the muon events are being recorded independently and without significant dead-time effects in the acquisition system.

4.5 P-32 SOURCE MEASUREMENT ATTEMPTS

A P-32 radioactive source was used to attempt calibration of the dEdx channel at lower energies in the beta electron range. The source was placed between the dEdx and bulk scintillator and data was collected over an extended period. However, upon analysis the collected histogram showed no statistically significant signal above the noise floor. Independent verification using a Geiger counter confirmed that the source activity had fallen below the detectable threshold meaning that the source had decayed to an unusable level over the period since its procurement, given P-32's relatively short half-life of 14.3 days. The P-32 data from this run was therefore not used for calibration.

CHAPTER 5

READOUT ELECTRONICS

5.1 OVERVIEW

The readout electronics chain for SPEED bridges the analogue signal output of the front-end electronics and the digital science data transmitted to the satellite's onboard computer. Once the front-end electronics have conditioned the PMT signal and the peak-hold detector has captured the pulse amplitude proportional to the deposited energy, the digitization, binning, and transmission of this data is handled entirely by firmware running on the microcontroller. The primary objective of the back-end digital electronics is to accumulate energy histograms for the dEdx and bulk detector channels in real time and package them as science data packets for downlink at regular intervals.

5.2 READOUT SIGNAL CHAIN

The complete signal chain from particle interaction to science packet transmission proceeds as follows. A charged particle entering the detector produces scintillation light in the dEdx or bulk scintillator, which is collected by the WLS fibres and guided to the corresponding PMT. The PMT converts the optical signal to a current pulse, which the front-end electronics amplify and shape through a series of analogue stages. The final stage of the front-end electronics is a peak-hold detector, which captures and holds the maximum voltage of the shaped pulse for long enough that the ADC on the microcontroller can sample it accurately. The digitized amplitude value is then processed by the firmware to determine which energy bin it belongs to, and the corresponding bin counter is incremented. Every three seconds, the accumulated histogram is packaged into a science packet and transmitted to the onboard computer over the SSI interface for downlink to the ground station. UART is used separately as a parallel debug interface during ground testing, allowing science packet contents to be verified.

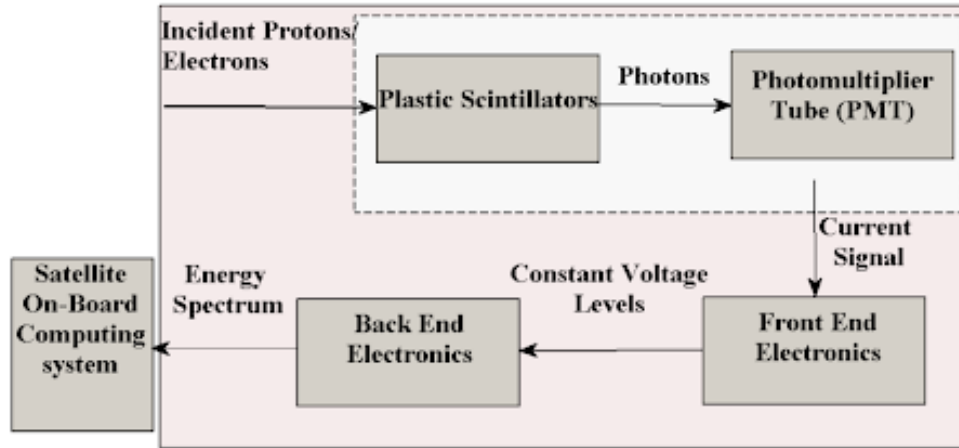


Figure 5.1: SPEED Signal Chain Simplified Diagram

5.3 MICROCONTROLLER FIRMWARE

We use Tiva TM4C123GH6PM [Texas Instruments (2014, 2013)] microcontroller in this setup. The firmware is written in C and runs on the microcontroller bare-metal, without an underlying operating system. All timing, data collection, and communication are managed through hardware peripheral configuration and interrupt-driven execution. The data collection pipeline is built around two timer interrupts operating at different cadences.

Table 5.1: Technical Specifications of the Tiva TM4C123GH6PM Microcontroller

Feature	Specification
Core	ARM Cortex-M4F with FPU
Clock Speed	80 MHz
Flash Memory	256 KB
SRAM	32 KB
EEPROM	2 KB
ADC	2x 12-bit (1 MSPS, 12 channels)
Timers	6x 16/32-bit, 6x 32/64-bit
PWM	16 Outputs (2 modules)
UART	8 Modules
I2C	4 Modules
SSI / SPI	4 Modules
CAN	2 Modules
USB	USB 2.0 Device/Host/OTG
GPIOs	Up to 43
Package	64-pin LQFP

The first timer interrupt fires every 0.1 seconds and serves as the primary data collection trigger. On each interrupt, the firmware reads the current ADC output which holds the peak voltage from the most recent particle event and maps it to the corresponding energy bin in either the dEdx or bulk histogram depending on which channel produced the signal. The second timer interrupt fires every 3 seconds and initiates a DMA transfer of the accumulated histogram data from the microcontroller's memory to the transmit buffer, from where it is formatted as a science packet and sent over the SSI interface to the onboard computer. The use of DMA for this transfer is important because it allows the bulk memory copy to proceed in hardware without occupying the CPU, ensuring that the 0.1-second sampling interrupt can continue firing without interruption during transmission.

5.4 KEY FIRMWARE CONTRIBUTIONS

5.4.1 Science Timer Fix

During firmware development, the science timer was found to be firing at an incorrect time intervals. Thoroughly reviewing the code identified the root cause as a data type mismatch in the `timerperiod` variable, which was declared a float value instead of a regular integer value (`uint64_t`). In C, assigning a float to an integer type silently results in value-based truncation rather than bitwise reinterpretation, causing the timer period to be set to a meaningless value and the interrupt to fire without any specific design. The consequence of this bug, had it gone undetected, would have been energy bins filled at the wrong rate producing energy histograms that appeared plausible but were scientifically meaningless. The fix required ensuring type consistency for the `timerperiod` variable throughout the timer configuration code.

5.4.2 HV Ramp Timer Extension

The high voltage supply to the PMTs must be ramped up slowly at startup to avoid damaging the dynode chain. This process requires a controlled ramp duration longer than 53 seconds. The ramp timing was implemented using a 32-bit hardware timer, but at the available peripheral clock frequency of 80MHz, a 32-bit counter overflows at approximately 54 seconds ($\frac{2^{32}}{80 \times 10^6} = 53.69$), imposing a hard ceiling on the maximum ramp duration achievable with a single timer period. The solution implemented was a software overflow counter in which a volatile `uint64_t` variable `rampTicks` is declared that is incremented each time the hardware timer fires an interrupt in the `HVRampTimerISR`. The true elapsed ramp time is then computed from the accumulated tick count rather than a single hardware period, effectively extending the ramp duration to an arbitrary length without requiring a different hardware timer or clock configuration.

5.4.3 Energy Binning and DMA Initialisation

The firmware accumulates energy data by sorting each ADC reading into one of a fixed number of bins in a histogram array, separately maintained for the dEdx and bulk channels. Each bin corresponds to a range of ADC values and therefore a range of deposited energies. The indexing logic for assigning ADC readings to the correct bin — taking into account the time-based filling and the separate bin arrays for each channel contained errors that caused readings to be assigned to incorrect positions in the

histogram. Additionally, the DMA transfer initialisation was fixed and its time-interval was validated to transfer data every 3 seconds. An incorrect DMA configuration is a leading cause of data corruption and makes the scientific analysis difficult.

5.4.4 ADC-to-Bin Mapping

A separate mapping error was identified in which the raw ADC readings from the dEdx and bulk channels were being routed to incorrect positions in the energy bin arrays, mostly in the lower energy bins. This is different from the indexing logic error described above because the mapping defines the relationship between a specific ADC value and the energy bin it represents, and an error here means that the shape of the recorded energy spectrum is distorted regardless of how correctly the histogram is otherwise managed. This bug is particularly dangerous because the resulting histograms contain plausible-looking distributions that would be difficult to identify as incorrect without cross-referencing against simulation. The correct mapping was fixed by carefully reviewing the ADC channel assignments and the bin boundary definitions.

CHAPTER 6

CONCLUSION

The work presented in this thesis covers three technical areas of the SPEED payload development, namely, Geant4 Monte Carlo simulation, laboratory data acquisition and analysis, and embedded firmware development for the readout electronics, each of which advances the payload towards flight readiness on the IITMSAT satellite.

The Geant4 simulation framework was extended to support a larger variety of radioactive decay chains and used to characterise the detector's response to cosmic muons and radioactive calibration sources: P-32 and Cs-137. The energy deposition distributions obtained from these simulations provided direct guidance on source placement strategies and the expected signal shapes in both the dEdx and bulk detector channels, establishing the simulation baseline against which experimental data could be compared. Noise modelling studies additionally quantified the dependence of detector performance on ambient gamma flux and WLS fibre reflectivity, establishing operational limits for reliable science data collection.

Laboratory calibration measurements were performed using the CAEN MC² Analyser with the physical payload replica. Cosmic muon data was successfully collected and analysed using a combined Gaussian and Landau fitting procedure, yielding the detector's count rate, most probable energy deposit, and timing resolution. The P-32 calibration run was unsuccessful due to source decay as confirmed by Geiger counter measurement

On the firmware side, several critical bugs were identified and resolved which include a data type mismatch causing incorrect science timer behaviour, a hardware timer range limitation overcome through a software overflow counter, an error in the energy bin indexing logic, and an incorrect ADC-to-energy-bin mapping. The end-to-end integration test required running the full chain from detector signal through the microcontroller to a verified science packet which provided reliable results on the replica model. Collectively, these contributions ensure that the firmware correctly and reliably accumulates energy histograms and transmits science packets, which is the fundamental requirement for the payload to produce valid science data in orbit.

The SPEED payload represents a genuine scientific instrument that will contribute real measurements to the study of Earth's radiation environment from orbit. The work presented here spans particle physics simulation, experimental detector characterisation, and low-level embedded firmware development which also reflects the breadth of technical knowledge required to bring such an instrument from design to flight readiness. Each contribution, right from a firmware bug fix to a simulation study, is a necessary piece of the larger system, and it is the integration of all these pieces that will ultimately determine the quality of the science data returned from orbit.

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